SOD-123 Surface Mount Schottky Barrier Rectifier

Features

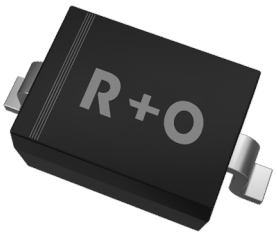
- V_R=40V/30/20V
- $I_{F(AV)}$=0.35A
- For use in low voltage, high frequency inverters
- Free wheeling, and polanty protection applications

Reverse Voltage
40/30/20 V
Forward Current
0.35 Ampere

Applications

The device is a single rectifier offering low VF and excellent high temperature stability.

SOD-123

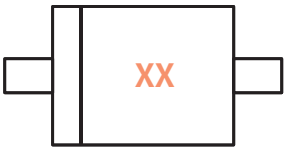


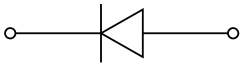
Mechanical Data

- Case: SOD-123
  Molding compound meets UL 94V-0 flammability rating, RoHS-compliant, halogen-free
- Terminals: Solder plated, solderable per MIL-STD-750,Method 2026
- Polarity: Cathode line denotes the cathode end

Function Diagram

Marking:
SD103AW:S4
SD103BW:S5
SD103CW:S6





Maximum Ratings (Ta=25°C Unless otherwise specified)

PARAMETER	SYMBOL	UNIT	SD103AW	SD103BW	SD103CW
Maximum repetitive peak reverse voltage	V_{RRM}	V	40	30	20
Maximum RMS voltage	V_{RMS}	V	28	21	18
Maximum DC blocking voltage	V_{DC}	V	40	30	20
Maximum Average Forward Rectified Current	$I_{F(AV)}$	mA	350		
Non-repetitive Peak Forward Surge Current @t=8.3ms Half-sine wave	I_{FSM}	A	2.0		
Power Dissipation	P_D	mW	400		
Junction temperature	T_j	°C	125		
Storage temperature range	T_{STG}	°C	-55-+150		
Typical thermal resistance	$R_{\theta J-A}$	°C /W	250		

Electrical Characteristics (Ta=25°C Unless otherwise noted)

PARAMETER	TEST CONDITIONS	SYMBOL	UNIT	SD103AW	SD103BW	SD103CW
Maximum forward voltage	I_F=20mA	V_F	V	0.37		
	I_F=200mA	V_F		0.60		
Maximum reverse current	V_R=30V	I_R	μA	5.0		
	V_R=20V					
	V_R=10V					
Capacitance between terminals	V_R = 0V, f = 1MHz	C_T	pF	50		
Reverse Recovery time	I_F=I_R=200mA I_{rr}=0.1×I_R, R_L=100Ω	t_{rr}	ns	10		

Ratings And Characteristics Curves (Ta=25°C Unless otherwise specified)

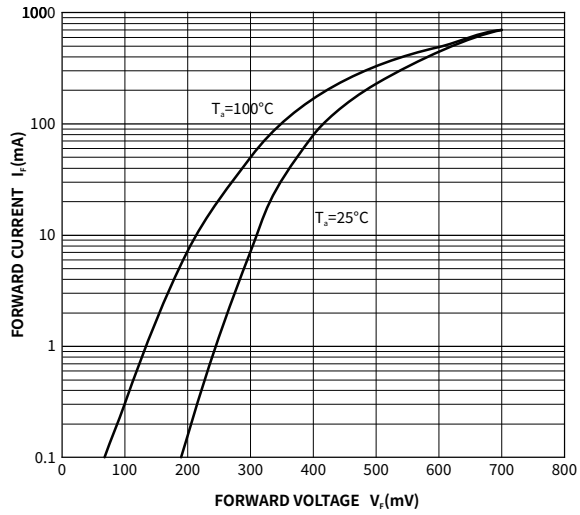


Fig.1 Typical Instaneous Forward Characteristics

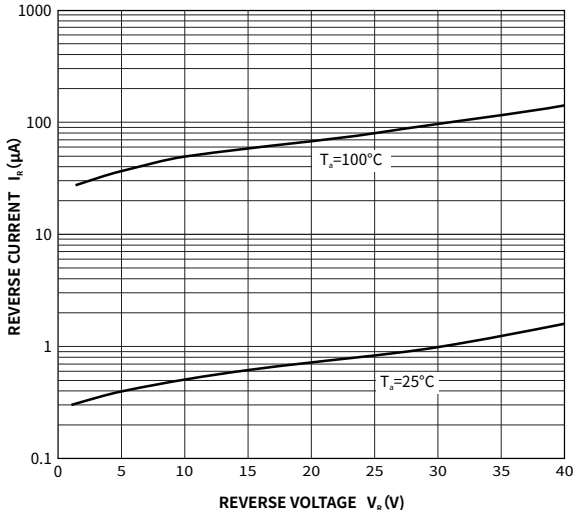


Fig.2 Typical Reverse Characteristics

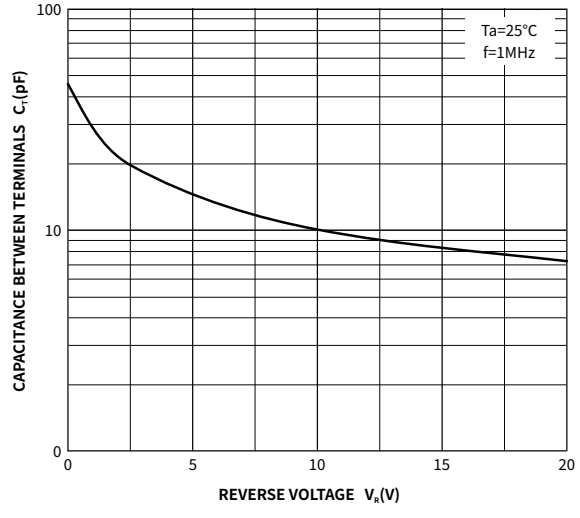


Fig.3 Typical Junction Capacitance

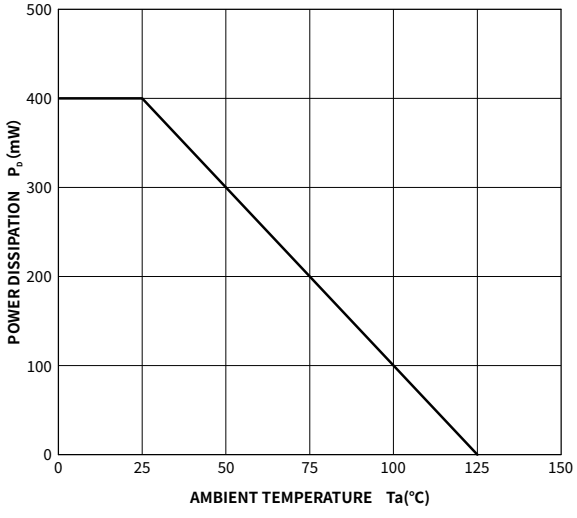
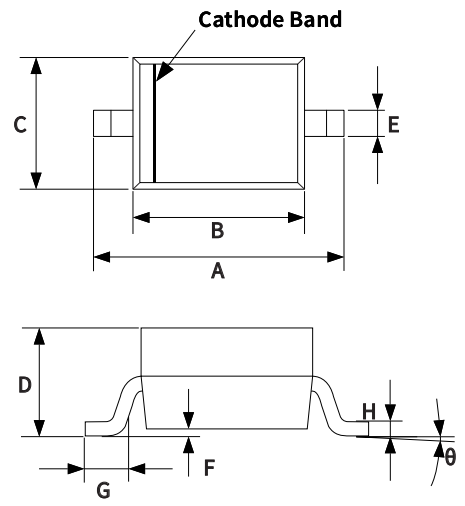


Fig.4 Power Derating Curve

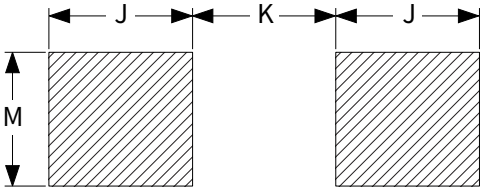
Ordering Information

PACKAGE	PACKAGE CODE	UNIT WEIGHT(g)	REEL(pcs)	BOX(pcs)	CARTON(pcs)	DELIVERY MODE
SOD-123	R1	0.012	3000	45000	180000	7''

Package Outline Dimensions (SOD-123)



Symbol	Dimensions			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
A	3.55	3.85	0.140	0.152
B	2.55	2.85	0.100	0.112
C	1.40	1.80	0.055	0.071
D	0.95	1.35	0.140	0.152
E	0.51	0.71	0.037	0.053
F	-	0.15	-	0.006
G	0.15	0.45	0.006	0.008
H	0.08	0.25	0.003	0.010
θ	-	8°	-	8°



Symbol	Dimensions			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
J	0.91	-	0.036	-
K	-	2.36	-	0.092
M	1.22	-	0.048	-